# Multi-Modal Edge Computing System

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## Abstract

Pharmaceutical manufacturing requires strict quality control to prevent defective vials from reaching patients. We present Argus, a multi-modal edge computing system for real-time detection of vial cracks, capping failures, and temperature anomalies on active production lines. By running computer vision and thermal sensing inference locally on embedded AI hardware, Argus achieves low-latency detection without centralized servers, reducing bandwidth and power consumption. Experimental results confirm high classification accuracy across defect categories, making Argus a scalable and cost-effective solution for zero-defect pharmaceutical production.

## Objectives

1. Build a real-time multi-modal edge system for pharmaceutical vial inspection.
2. Detect cracks and improper caps using edge-based computer vision.
3. Monitor vial temperature for regulatory compliance.
4. Minimize latency, power, and bandwidth via local edge inference.
5. Reduce defective product waste through automated quality control.
6. Design a scalable architecture for industrial deployment.

## Materials and Methods

**Hardware**

- Edge device (Raspberry Pi / Jetson)
- Halo AI Chip
- ArduCam 60MP Industrial camera module
- MLX90614 Infrared sensor

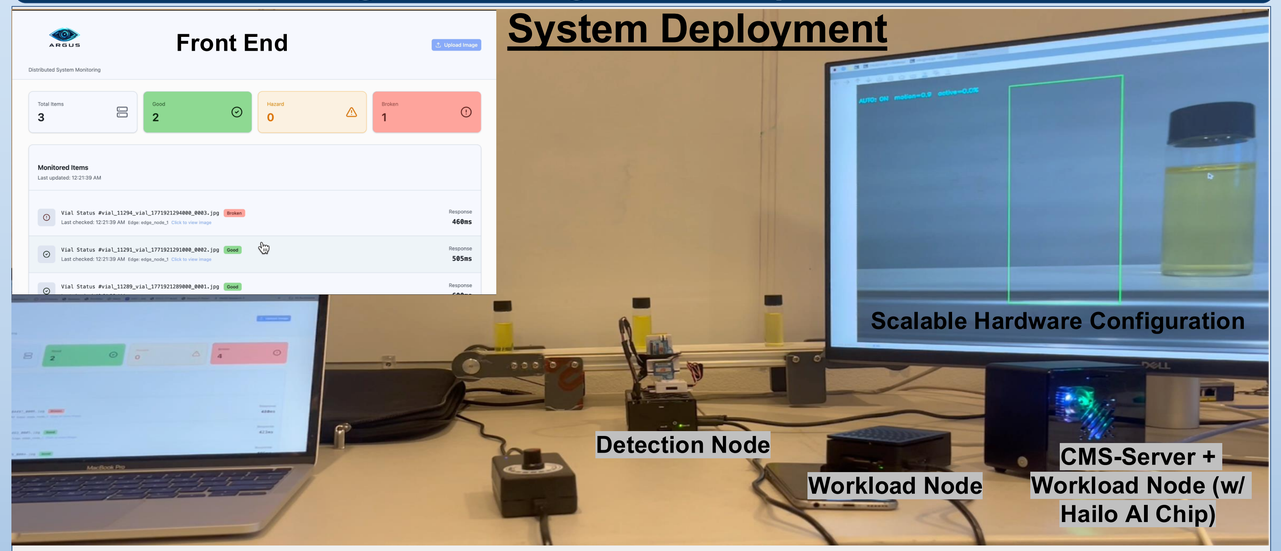
**Software**

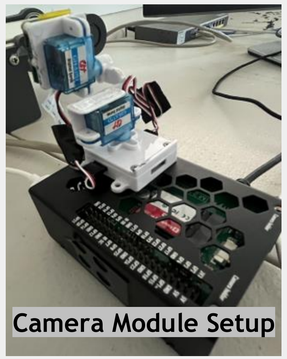
- YOLO-based classification model
- Communication Protocols: TCP, I2C
- Least Loaded First Scheduling
- User Interface Monitoring System

**Method**

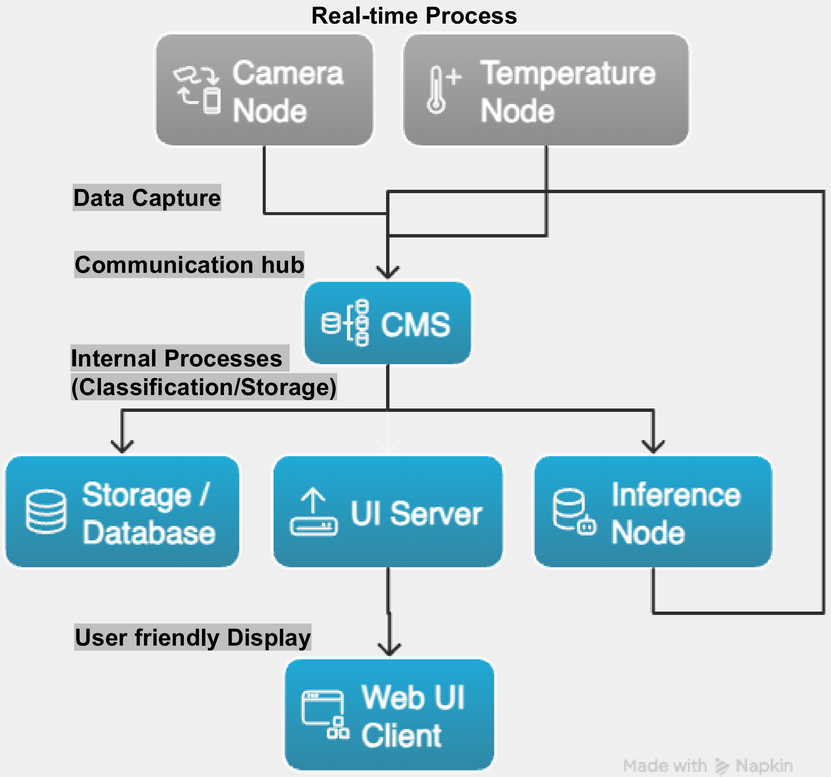
1. Capture vial data (image/temp)
2. Perform real-time defect classification
3. Log anomalies and trigger UI alerts

## Diagrams/Figures/Experiments



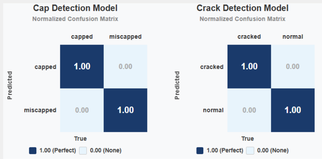



### Workflow Visualization




## Results

**Model Validation Results**




**Experimental Test Results**

| Category | Total | Correct | Incorrect | Model Accuracy |
|---|---|---|---|---|
| Cracked | 113 | 111 | 2 | **98.2%** |
| Normal | 121 | 111 | 10 | **91.7%** |

## Additional Information

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## References

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[4] M. F. Habte, B. A. Tegegne, and T. Y. Alemayehu, "Anesthetics drug wastage and preventive strategies: Systematic review," *Journal Name*, Year. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11253927/